Tourism and Recreation EWG 3.1

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1 Tourism and Recreation EWG 3.1

1.1 About the TR Goal

This goal captures the quantity, quality, and sustainability of experiences that people have when visiting coastal and marine areas.

CORE DEFINITION

- ***Focus:*** Value that people derive from experiencing and enjoying coastal areas
- ***Not about:*** Revenue or livelihoods generated by tourism (covered under Livelihoods and Economies)
- ***Emphasis:*** Personal and cultural value of coastal recreation experiences

1.2 Goal model

The overall Tourism & Recreation score is the average of three components:

TR = average(Quantity, Quality, Sustainability)

Component	What it measures
Quantity	Tourist arrivals (average of domestic and international visitors from SECTUR DataTur)
Quality	Average of safety of tourists (crime index) and availability of activities (food & drink, culture & gulf related, and other recreation relative to tourist arrivals)
Sustainability	Average of hotel certifications (Distintivo S, Green Key, Green Globe), Blue Flag beaches, tourism management investment (CONANP programs), and job quality in tourism

Scores are calculated for each of the nine OHI Gulf of California regions from 2019 to 2024. A score of 100 represents the reference point (ideal sustainable state).

1.3 Overall Tourism and Recreation status

The overall Tourism and Recreation (TR) current status score summarizes how close each OHI Gulf of California region is to sustainable coastal recreation experiences. It integrates three equally weighted components: **Quantity** (visitor arrivals), **Quality** (safety and activity availability), and **Sustainability** (certifications, beach management, job quality, and public investment). Scores are calculated annually from 2019 to 2024 for all nine marine regions. A score of 100 represents the reference point (ideal sustainable state).

What scores mean: Higher TR scores signal that a region is closer to improved recreation conditions across visitation levels according to government targets, visitor experience quality, and long-term ability to manage tourism sustainably. Lower scores highlight where arrivals, safety, activity availability, certifications, employment quality, or public investment still fall short of reference targets. Managers can use regional lines to compare trajectories; communities can see whether gains in visitor arrivals are matched by improvements in safety and sustainability.

Data sources: Merged current status outputs from SECTUR DataTur (quantity), SESNSP and INEGI DENUE (quality), and CONANP, FEE Blue Flag, SECTUR certifications, and INEGI Censos Economicos (sustainability).

Method: For each region and year, the three component scores are averaged. Component scores themselves are built from sub-indicators described in the sections below.

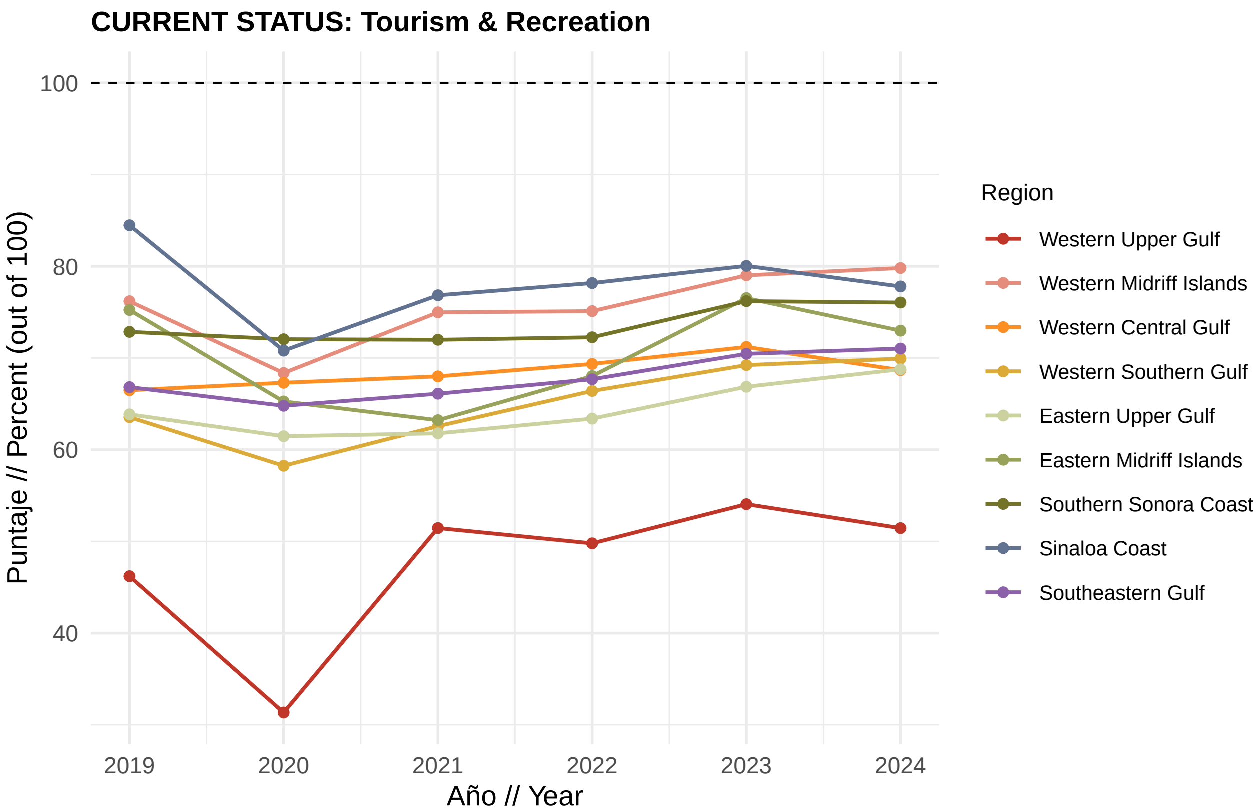
$$\mathrm{TR}_{r,t} = \frac{Q_{r,t} + Ql_{r,t} + S_{r,t}}{3}$$

where Q = Quantity, Ql = Quality, and S = Sustainability (each already expressed on a 0 to 100 scale, so TR is also on a 0 to 100 scale). Scores above 100 are not applied at this level because component scores are capped where noted below.

Things to note: Crime-based quality indicators have limited SESNSP coverage for Regions 2 and 6 (Western and Eastern Midriff Islands). Activity availability depends on DENUE establishment counts within 50 km of the coast and SECTUR/DENUE gap-filled arrivals. Sustainability job-quality scores use INEGI census years (2003 to 2023) interpolated to annual values for 2019 to 2024.

1.3.1 Regional trajectories for overall TR

The chart below shows the overall TR current status score by region and year. Each colored line is one OHI region; compare values across years. The dashed line at 100 marks the reference goal. Values below 100 indicate that one or more components remain below their reference targets; values at or near 100 indicate progress toward sustainable tourism and recreation across quantity, quality, and sustainability dimensions. Steeper upward slopes suggest faster improvement; flat or declining lines may warrant closer review of the underlying sub-indicators.



1.4 Quantity

The Quantity component measures the scale of coastal and marine visitor activity relative to Plan Mexico 2030 tourism growth targets. It reflects how close regional tourist arrivals are to planned increases in domestic and international visitation, rather than measuring revenue or economic impact (those are covered under Livelihoods and Economies).

What scores mean: A score of 100 means arrivals have reached or exceeded the 2030 Plan Mexico target for that arrival type. Scores below 100 indicate visitation still below target; stakeholders planning infrastructure or marketing can use these gaps to prioritize regions where growth lags policy goals.

Data sources: SECTUR DataTur hotel monitoring reports (`llegadas de turistas residentes` for domestic arrivals and `llegadas de turistas no residentes` for international arrivals), aggregated to OHI regions. Missing municipal arrival data are gap-filled using DENUE accommodation employee models and state-level domestic/international arrival proportions.

Data processing: Tourist centers are geocoded and intersected with OHI region boundaries. Municipal gaps are filled by scaling DENUE accommodation employment to estimated arrivals using state-level domestic/international splits. Arrivals are summed to the regional level for 2019 to 2024.

Method: Quantity is the simple average of domestic and international arrival scores. Each arrival type is compared to a fixed 2030 target based on 2024 baseline arrivals. Scores above 100 are capped at 100.

$$\text{Quantity}_{r,t} = \frac{\text{Domestic}_{r,t} + \text{International}_{r,t}}{2}$$

Data limitations: Regions without direct SECTUR tourist-center reporting rely on DENUE gap-filling, which introduces additional uncertainty. San Ignacio has limited DataTur coverage. International and domestic targets use separate 2030 growth rates (30% and 9.8% above 2024, respectively).

1.4.1 Quantity component result

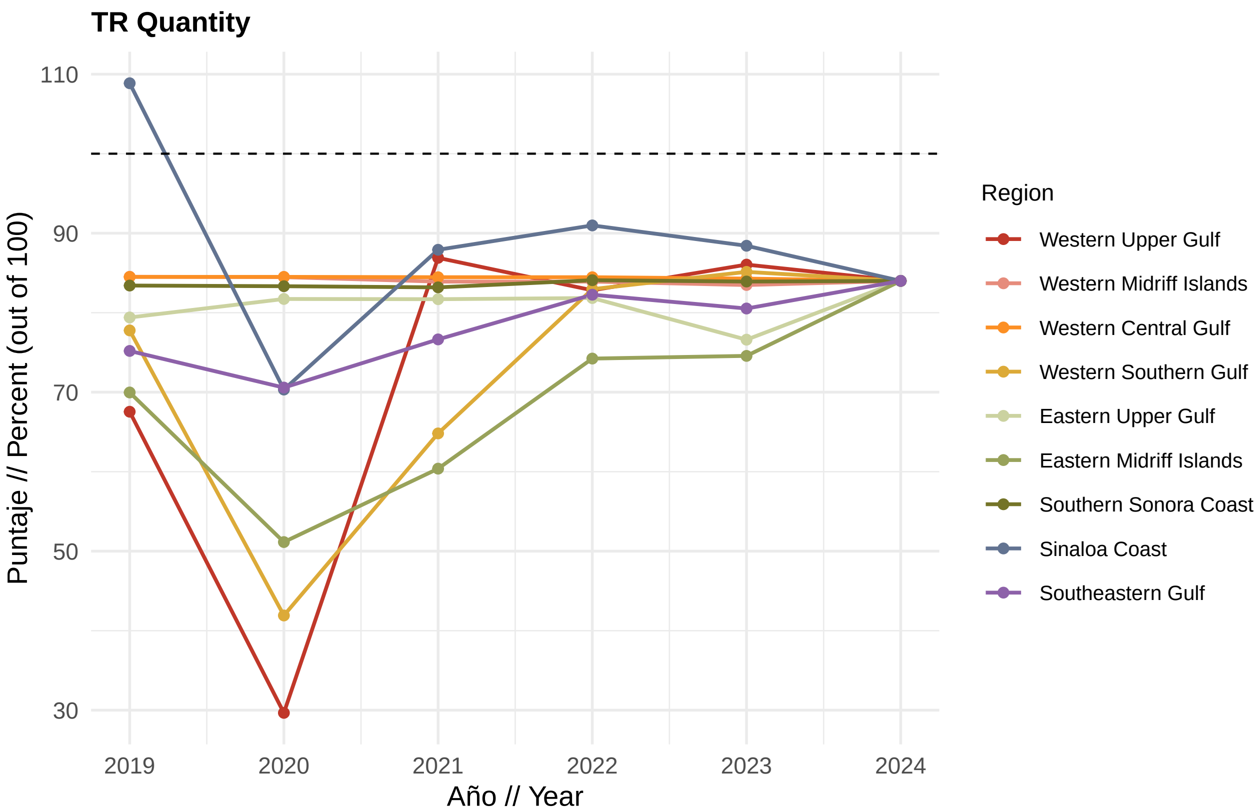
What it measures: Combined progress toward Plan Mexico 2030 arrival targets for domestic and international visitors.

Data source: SECTUR DataTur (2019 to 2024), with DENUE gap-filling for municipalities without direct reporting.

Method: Average of capped domestic and international current status scores per region and year.

$$\text{Quantity}_{r,t} = \frac{1}{2}\left(\min\left(100,\ \frac{A^{\text{dom}}_{r,t}}{A^{\text{dom}}_{r,2024} \times 1.098} \times 100\right) + \min\left(100,\ \frac{A^{\text{int}}_{r,t}}{A^{\text{int}}_{r,2024} \times 1.30} \times 100\right)\right)$$

where A^{dom} and A^{int} are regional domestic and international tourist arrivals, and the denominators are the 2030 targets (9.8% above 2024 for domestic, 30% above 2024 for international).



1.4.1.1 Quantity subcomponents

1.4.1.1.1 Domestic arrivals

What it measures: How close regional domestic tourist arrivals are to the Plan Mexico 2030 target of 9.8% growth above 2024 levels.

What scores mean: Higher scores indicate domestic visitation is approaching or meeting national growth targets. Lower scores suggest domestic tourism recovery or expansion is lagging, which may affect local businesses reliant on national visitors.

Data source: SECTUR DataTur `llegadas de turistas residentes`, assigned to OHI regions via tourist center geocoding and spatial intersection.

Data processing: Monthly DataTur hotel reports are cleaned, pivoted to annual totals, and assigned to regions. Municipalities without arrival data are gapfilled using estimated arrivals based on the number of hotel employees.

Method: Current status is the ratio of observed domestic arrivals to the 2030 target, which is a 9.8% increase by 2030, expressed as a percentage and capped at 100.

$$\text{Domestic}_{r,t} = \min\left(100,\ \frac{A_{r,t}^{\text{dom}}}{A_{r,2024}^{\text{dom}} \times 1.098} \times 100\right)$$

1.4.1.1.2 International arrivals

What it measures: How close regional international tourist arrivals are to the Plan Mexico 2030 target of 30% growth above 2024 levels.

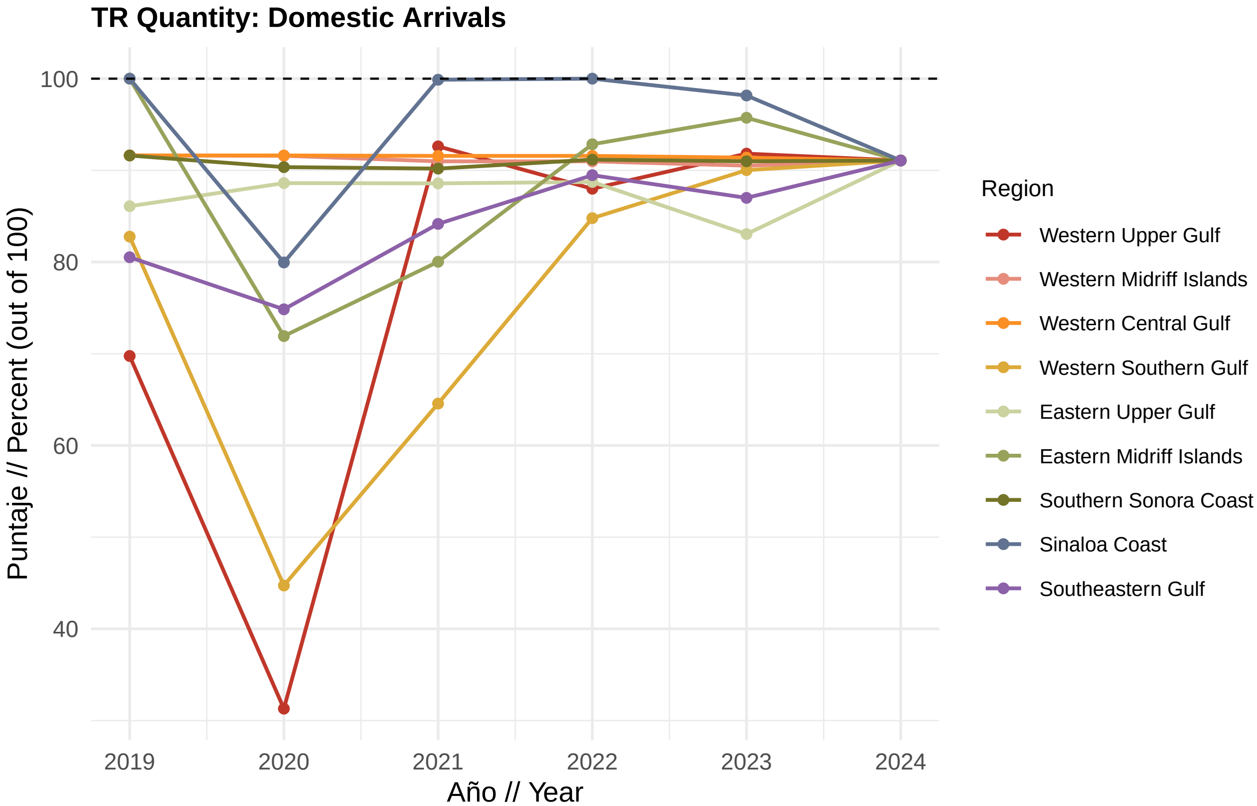
What scores mean: Higher scores reflect stronger international demand relative to policy targets. Lower scores may signal barriers to international visitation such as connectivity, marketing, or perceived destination quality.

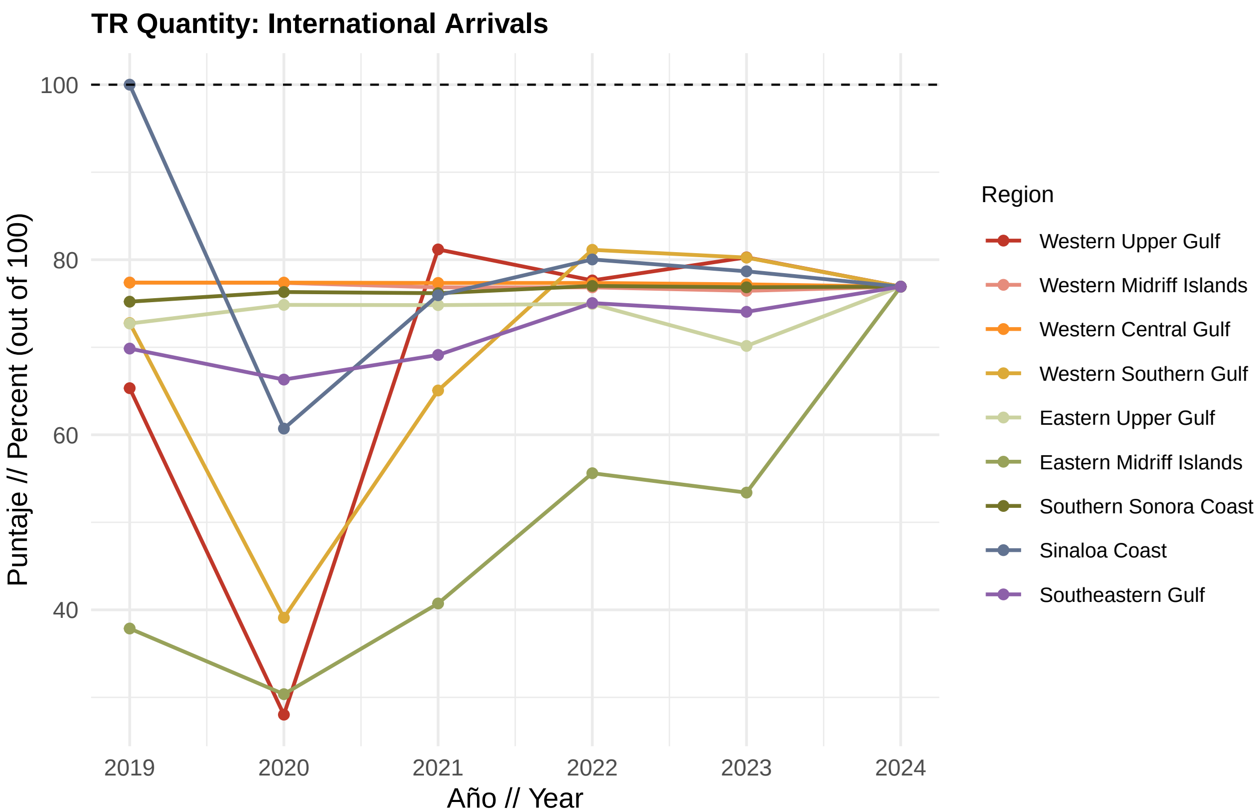
Data source: SECTUR DataTur `llegadas de turistas no residentes`, assigned to OHI regions via tourist center geocoding and spatial intersection.

Data processing: Same pipeline as domestic arrivals, with separate state-level international arrival proportions for gap-filling.

Method: Current status is the ratio of observed international arrivals to the 2030 target, which is a 30% increase by 2030, expressed as a percentage and capped at 100.

$$\text{International}_{r,t} = \min\left(100,\ \frac{A_{r,t}^{\text{int}}}{A_{r,2024}^{\text{int}} \times 1.30} \times 100\right)$$


TR Quantity: Domestic Arrivals
Puntaje // Percent (out of 100)
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf


TR Quantity: International Arrivals
Puntaje // Percent (out of 100)
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf

1.5 Quality

The Quality component captures how safe and enriching coastal visits are for tourists. It combines two equally weighted sub-indicators: **safety of tourists** (inverse crime index) and **availability of activities** (food and beverage, culture and Gulf-related, and other recreation services relative to tourist arrivals).

What scores mean: Higher Quality scores indicate safer destinations with more recreation services per visitor. Lower scores may reflect elevated crime rates, fewer dining and cultural options relative to arrivals, or both. Destination managers can use sub-indicator charts to see whether safety or activity availability drives regional differences.

Data sources: SESNSP municipal crime statistics (weighted by crime type), INEGI municipal population (interpolated), INEGI DENUE economic units within 50 km of the coast, and SECTUR/DENUE gap-filled tourist arrivals.

Data processing: Crime counts are filtered to tourism-relevant subtypes, weighted, and divided by interpolated coastal municipal population. DENUE establishments are classified into three activity categories; employee counts are estimated from establishment size classes and divided by regional tourist arrivals. Activity scores use the 90th quantile of employees-per-arrival across all region-year observations as the reference (cross-regional comparison, not within-region).

Method: Quality is the average of the crime safety score and the weighted activities score.

$$\text{Quality}_{r,t} = \frac{\text{Crime}_{r,t} + \text{Activities}_{r,t}}{2}$$

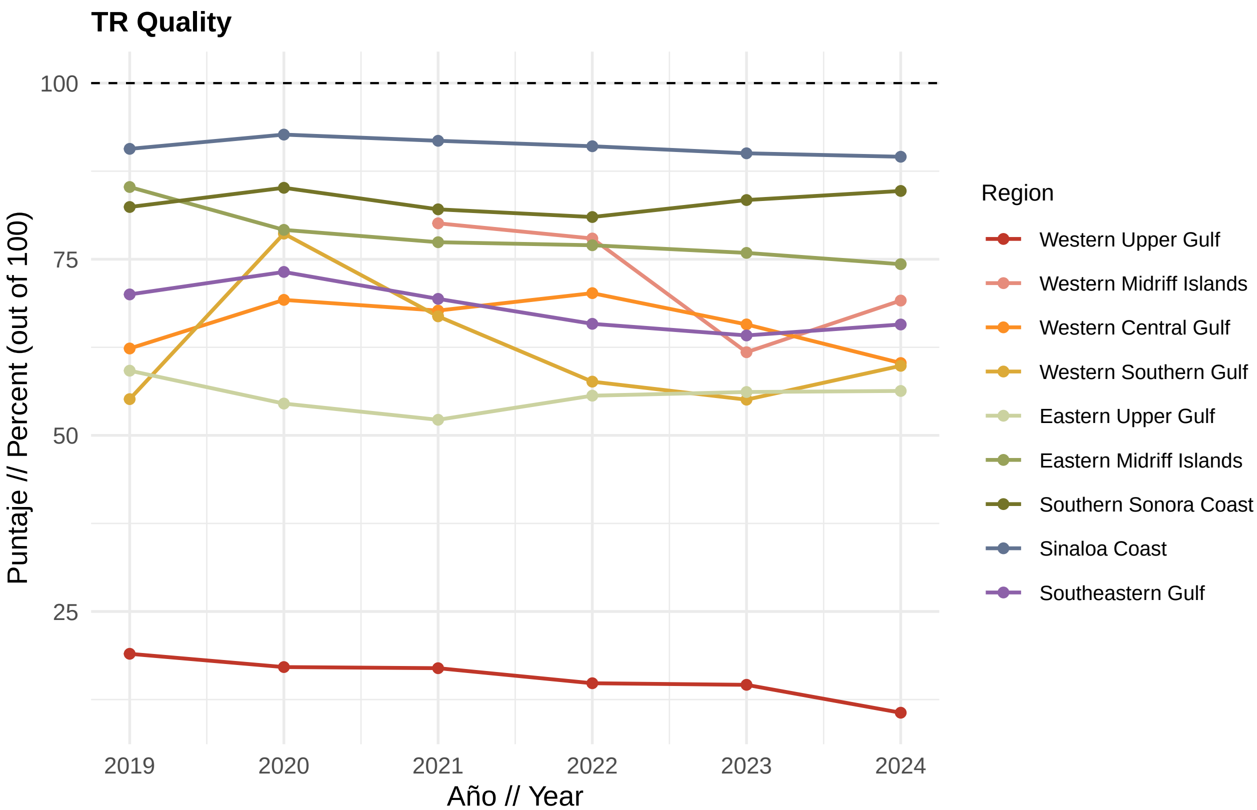
1.5.1 Quality component result

What it measures: Combined tourist safety and availability of recreation services relative to reference conditions.

Data source: SESNSP, INEGI population, INEGI DENUE, and SECTUR arrivals.

Method: Simple average of crime safety and activities scores.

$$\text{Quality}_{r,t} = \frac{\text{Crime}_{r,t} + \text{Activities}_{r,t}}{2}$$



1.5.1.1 Quality subcomponents

1.5.1.1.1 Safety of tourists (crime index)

What it measures: Relative safety from crimes that plausibly alter safety, cleanliness, aesthetics, or atmosphere in spaces tourists visit, which in turn affects visit quality.

What scores mean: A score of 100 means zero weighted crimes per coastal inhabitant (the best possible outcome). A score of 0 means the region-year has the highest weighted crime rate observed across all Gulf regions and years. Intermediate values reflect proportional distance from the worst-case rate. For visitors and operators, higher scores indicate lower risk environments for tourists because of a lower crime rate.

Data source: SESNSP (Secretariado Ejecutivo del Sistema Nacional de Seguridad Publica) annual municipal crime incidence by crime subtype; INEGI population interpolated to annual estimates for Gulf-bordering municipalities. **Mexicali is excluded** from crime calculations because its crime profile reflects border dynamics rather than Gulf coastal tourism.

Selection rule: Crimes are included if they plausibly alter safety, cleanliness, aesthetics, or atmosphere in spaces tourists visit. Subtypes not meeting this criterion (e.g., crimes committed by public servants, prison escapes) are dropped.

Two weight categories:

Category	Weight	Rationale
Crimes against person	2	Direct threat to visitor safety or strong link to unsafe atmosphere
Crimes against property	1	Financial loss or environmental damage affecting visit quality

Crimes against person (weight = 2):

- Sexual crimes: Abuso sexual, Acoso sexual, Hostigamiento sexual, Otros delitos que atentan contra la libertad y la seguridad sexual, Violacion equiparada, Violacion simple
- Violence and bodily harm: Feminicidio, Homicidio doloso, Lesiones dolosas, Otros delitos que atentan contra la vida y la integridad corporal
- Deprivation of liberty: Otros delitos que atentan contra la libertad personal, Rapto, Secuestro, Trafico de menores
- Coercion and threats: Amenazas, Extorsion, Violencia de genero en todas sus modalidades distinta a la violencia familiar
- Robbery with personal exposure: Robo a transeunte en espacio abierto al publico, Robo a transeunte en via publica, Robo en transporte individual, Robo en transporte publico colectivo, Robo en transporte publico individual
- Society-linked crimes: Corrupcion de menores, Otros delitos contra la sociedad, Trata de personas, Narcomenudeo

Crimes against property (weight = 1):

- Dano a la propiedad, Fraude, Otros delitos contra el patrimonio, Otros robos, Robo a casa habitacion, Robo a institucion bancaria, Robo a negocio, Robo a transportista, Robo de autopartes, Robo de vehiculo automotor, Contra el medio ambiente

Weighted crimes per coastal inhabitant:

$$C_{r,t} = \frac{\sum_i w_i \cdot n_{i,r,t}}{P_{r,t}^{\text{coastal}}}$$

where $n_{i,r,t}$ is the count of crime subtype i in region r and year t, w_i is the category weight (2 or 1), and $P_{r,t}^{\text{coastal}}$ is interpolated coastal municipal population summed to the region.

Method: Weighted crimes per capita are converted to a safety score where the region-year with the highest per capita crime rate receives 0 and zero crimes receives 100. Scores above 100 are capped at 100.

$$\text{Crime}_{r,t} = \min\left(100,\ \left(1 - \frac{C_{r,t}}{C_{\max}}\right) \times 100\right)$$

where $C_{\max}$ is the maximum weighted crimes per coastal inhabitant across all regions and years.

Data limitations: Regions 2 and 6 (Western and Eastern Midriff Islands) may have limited or no documented SESNSP crime data in the municipal records used here, which can produce missing or uncertain safety scores for those regions. Population for Region 2 uses special handling because San Quintin municipality was created in 2020.

1.5.1.1.2 Availability of activities

What it measures: Availability of food and beverage, culture and Gulf-related, and other recreation services relative to tourist arrivals, using estimated DENUE employees per tourist arrival.

What scores mean: Higher scores mean a region offers more recreation employees per visitor compared to the Gulf-wide 90th quantile benchmark. Lower scores suggest thinner service provision relative to visitation, which may affect visitor satisfaction even when arrival numbers are high.

Data source: INEGI DENUE service establishment shapefiles (food and beverage, cultural and recreational services) within 50 km of the Gulf coast; gap-filled SECTUR tourist arrivals.

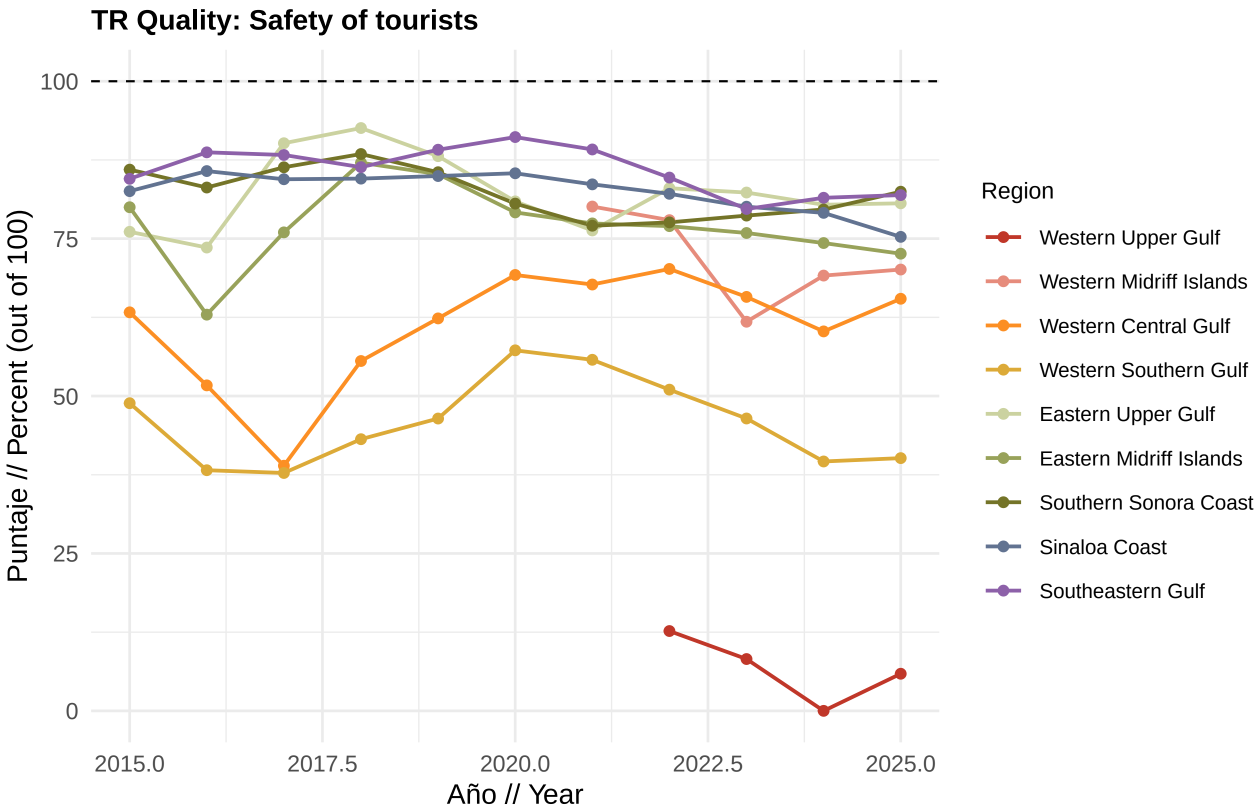
Data processing: Establishments are assigned employee counts from DENUE size-class midpoints, summed by category and region-year, then divided by tourist arrivals. The 90th quantile of employees-per-arrival is computed across **all region-year observations** (cross-regional, not within-region).

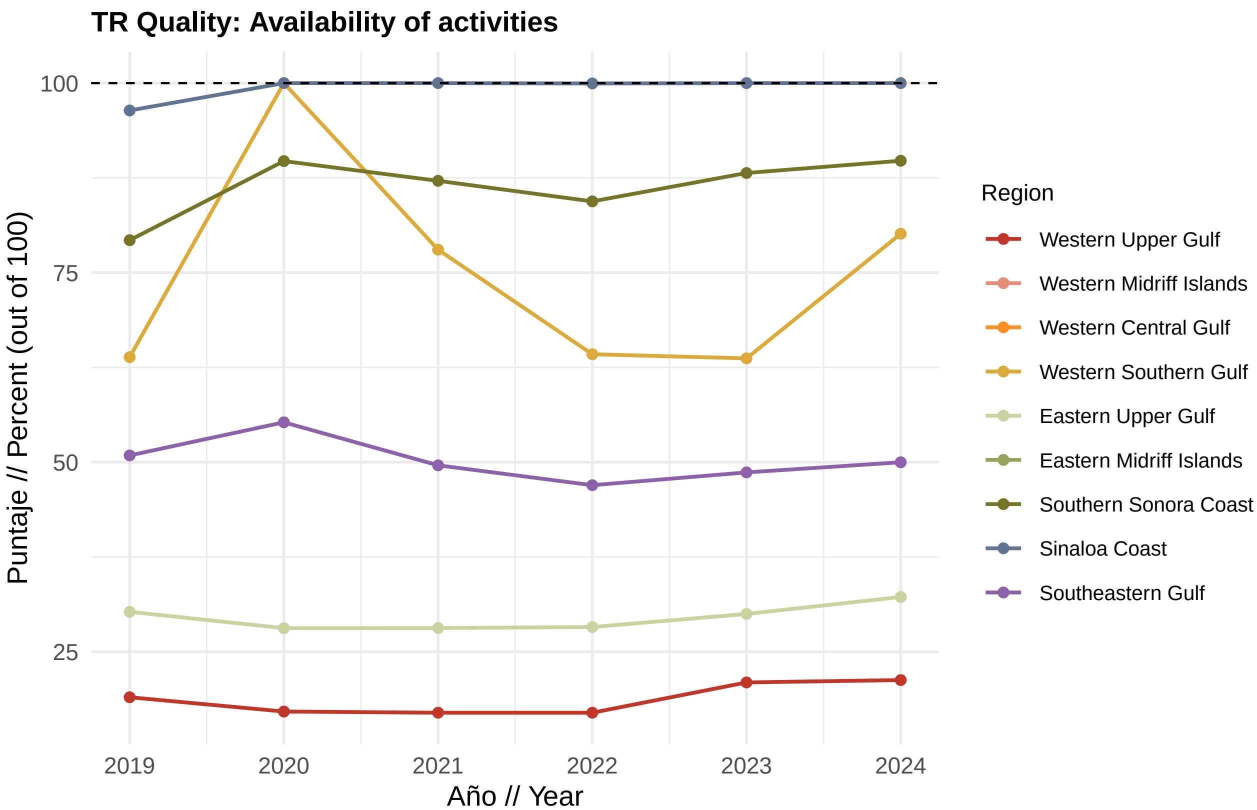
Method: Three activity categories are scored against the Gulf-wide 90th quantile reference, then combined with weights of 2/5, 2/5, and 1/5. Scores above 100 are capped at 100.

$$\text{Activities}_{r,t} = \frac{2}{5}\,\text{Food\&Bev}_{r,t} + \frac{2}{5}\,\text{Culture}_{r,t} + \frac{1}{5}\,\text{Other}_{r,t}$$

$$\text{Category}_{r,t} = \min\left(100,\ \frac{(E_{r,t}/A_{r,t})}{Q_{90}(E/A \text{ across all } r,t)} \times 100\right)$$

where $E_{r,t}$ is category employees, $A_{r,t}$ is tourist arrivals, and Q_{90} is the 90th quantile of the employees-per-arrival ratio pooled across all regions and years for that category.





1.5.1.1.3 Quality activity categories

Food and beverage activities

What it measures: Density of food and beverage service employees relative to tourist arrivals, compared to the 90th quantile across all region-year observations.

What scores mean: Scores near 100 indicate among the highest food and beverage service density per visitor in the Gulf. Lower scores suggest fewer dining options per arrival relative to top-performing region-years.

Data source: INEGI DENUE `servicios de alojamiento temporal y preparacion de alimentos y bebidas` establishments within 50 km; SECTUR/DENUE arrivals.

Data processing: Employee counts estimated from DENUE establishment size classes; summed by region and year; divided by gap-filled arrivals. Reference is Q_{90} of employees-per-arrival pooled across all regions and years.

Method: Employees per tourist arrival in year t divided by the Gulf-wide 90th quantile reference, capped at 100.

$$\text{Food\&Bev}_{r,t} = \min\left(100,\ \frac{(E^{\text{fb}}_{r,t}/A_{r,t})}{Q_{90}(E^{\text{fb}}/A \text{ across all } r,t)} \times 100\right)$$

Culture and Gulf activities

What it measures: Density of cultural, heritage, marine tourism, and museum service employees relative to tourist arrivals, compared to the 90th quantile across all region-year observations.

What scores mean: Higher scores reflect richer cultural and Gulf-oriented recreation per visitor. Lower scores may indicate fewer museums, heritage sites, or marine tourism services relative to visitation.

Data source: INEGI DENUE cultural and recreational service establishments (museums, historical sites, marine tourism, nature heritage) within 50 km; SECTUR/DENUE arrivals.

Data processing: Same quantile-based scoring as food and beverage, using culture-specific DENUE activity codes.

Method: Employees per tourist arrival in year t divided by the Gulf-wide 90th quantile reference, capped at 100.

$$\text{Culture}_{r,t} = \min\left(100,\ \frac{(E^{\text{cult}}_{r,t}/A_{r,t})}{Q_{90}(E^{\text{cult}}/A \text{ across all } r,t)} \times 100\right)$$

Other recreation activities

What it measures: Density of other recreational service employees (fitness centers, sports clubs, entertainment venues, etc.) relative to tourist arrivals, compared to the 90th quantile across all region-year observations.

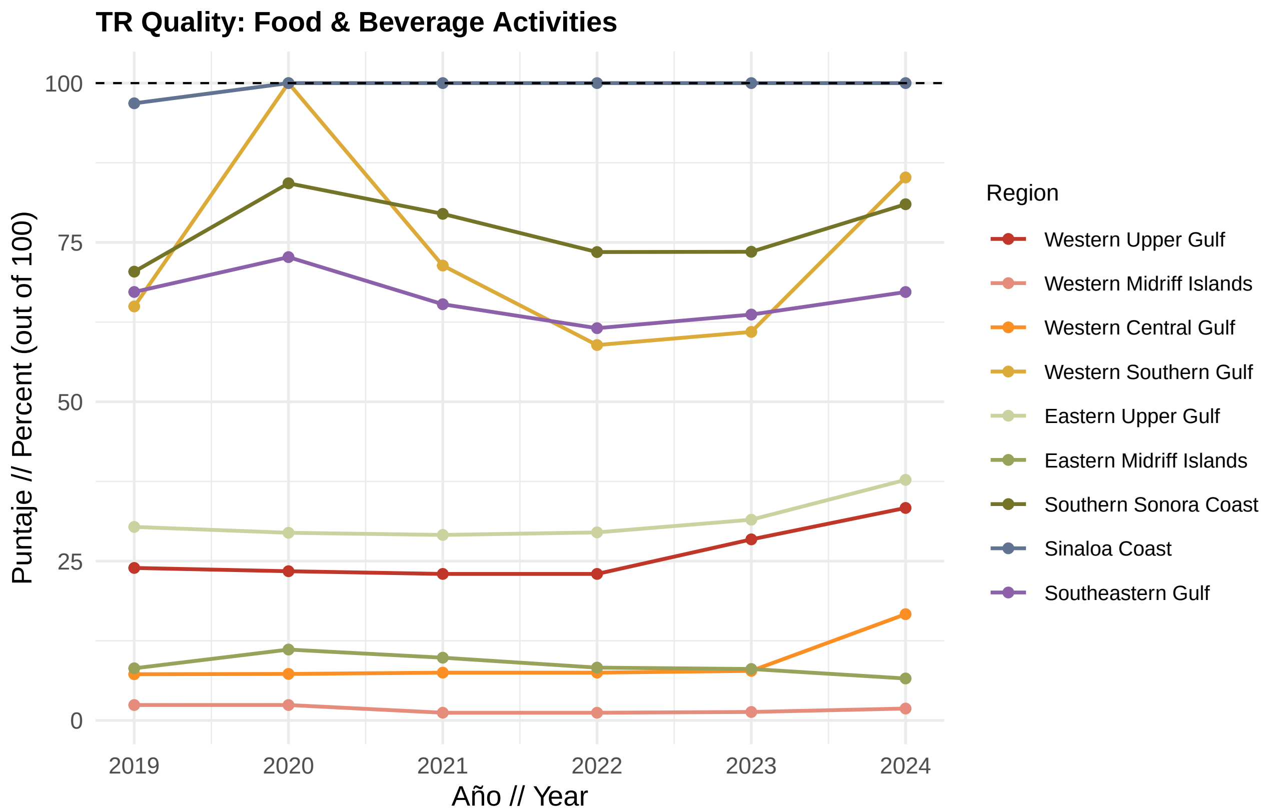
What scores mean: Higher scores indicate more general recreation infrastructure per visitor. Lower scores suggest limited supplementary recreation beyond core dining and cultural offerings.

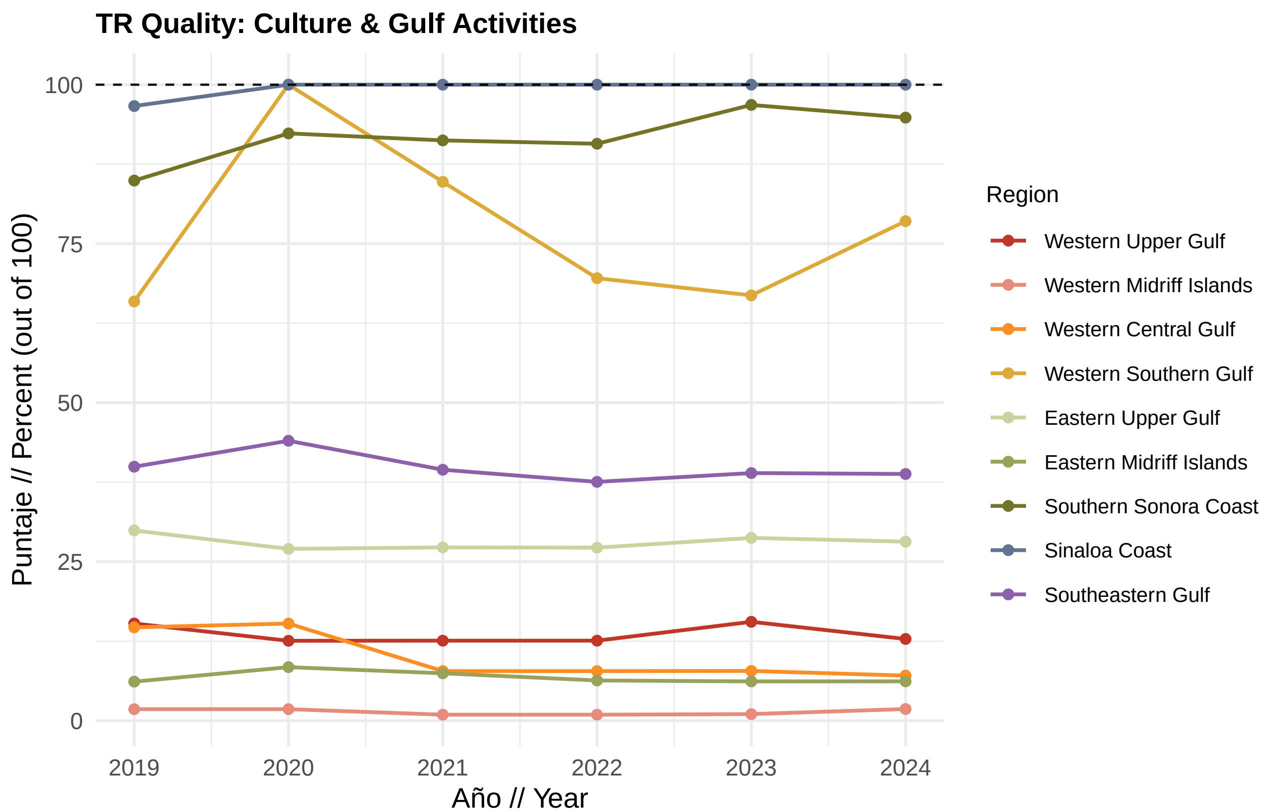
Data source: INEGI DENUE other recreational service establishments within 50 km; SECTUR/DENUE arrivals.

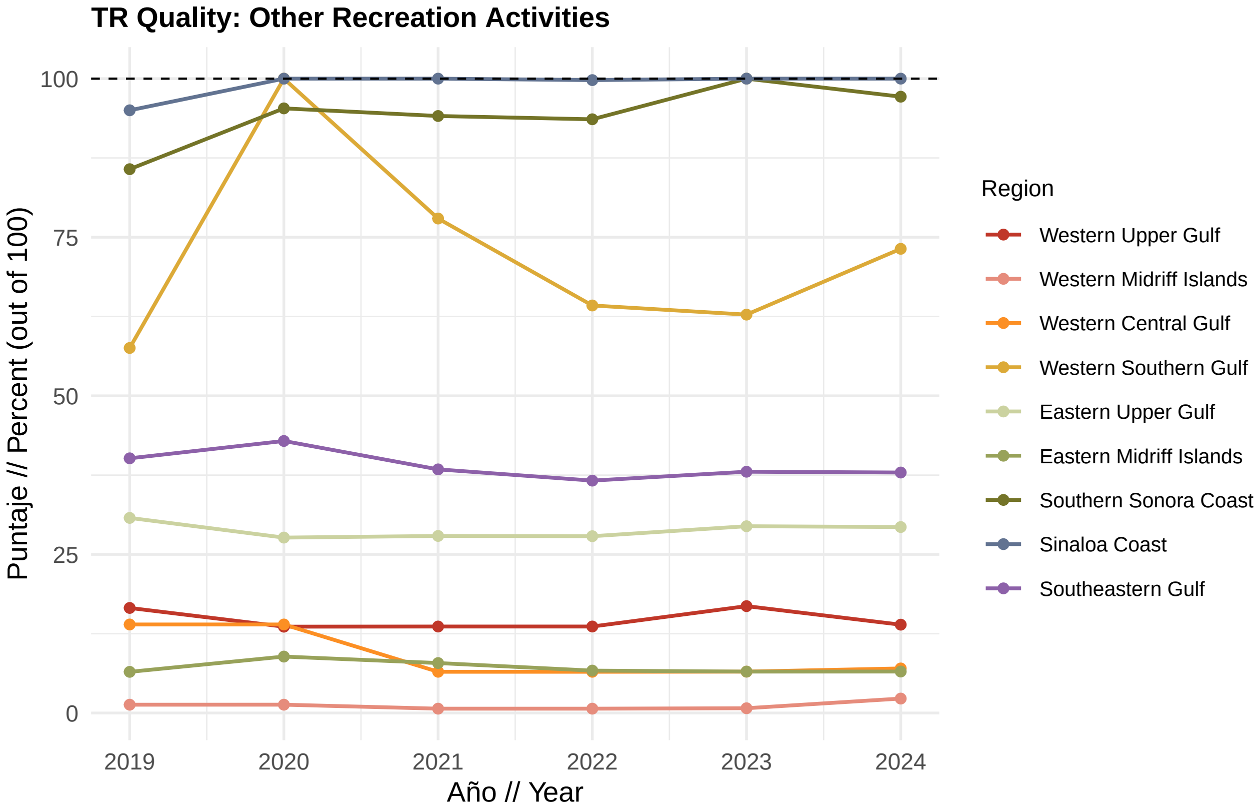
Data processing: Same quantile-based scoring pipeline; this category receives 1/5 weight in the merged activities score.

Method: Employees per tourist arrival in year t divided by the Gulf-wide 90th quantile reference, capped at 100.

$$\text{Other}_{r,t} = \min\left(100,\ \frac{(E^{\text{oth}}_{r,t}/A_{r,t})}{Q_{90}(E^{\text{oth}}/A \text{ across all } r,t)} \times 100\right)$$


TR Quality: Food & Beverage Activities
Puntaje // Percent (out of 100)
100
75
50
25
0
2019
2020
2021
2022
2023
2024
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf


TR Quality: Culture & Gulf Activities
Puntaje // Percent (out of 100)
100
75
50
25
0
2019
2020
2021
2022
2023
2024
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf



1.6 Sustainability

The Sustainability component evaluates whether tourism growth is supported by certified sustainable practices, well-managed beaches, equitable formal employment, and adequate potential public investment in coastal conservation and protected area management.

What scores mean: Higher Sustainability scores indicate stronger adoption of green certifications, better beach management, improved tourism job quality, and sustained public investment in protected areas. Lower scores highlight gaps in sustainable tourism infrastructure that may undermine long-term coastal health even when visitor numbers grow.

Data sources: SECTUR Distintivo S, Green Key, and Green Globe certification records matched to INEGI DENUE hotels; FEE Blue Flag Mexico beach registry; INEGI Censos Economicos (marine tourism employment, census years 2003, 2008, 2013, 2018, 2023); CONANP federal budget program data and SHCP fee return records (NOSSA).

Data processing: Hotel certifications are crosswalked to DENUE accommodation units with 100+ employees within 50 km. Blue Flag beach length is measured against total urban tourist beach length per region. Job quality metrics use constant 2023 pesos and coastal population multipliers, then interpolate census-year scores to annual values for 2019 to 2024. Investment programs compare current CONANP budgets to 2012 to 2018 averages in constant 2026 pesos.

Method: Sustainability is the average of four sub-components: certifications (hotel and Blue Flag), job quality, and tourism management investment.

$$\text{Sustainability}_{r,t} = \frac{\text{Cert}_{r,t} + \text{JobQual}_{r,t} + \text{Invest}_{r,t}}{3}$$

Note: Certifications merges hotel sustainability certifications and Blue Flag beaches before entering the overall sustainability average alongside job quality and investment.

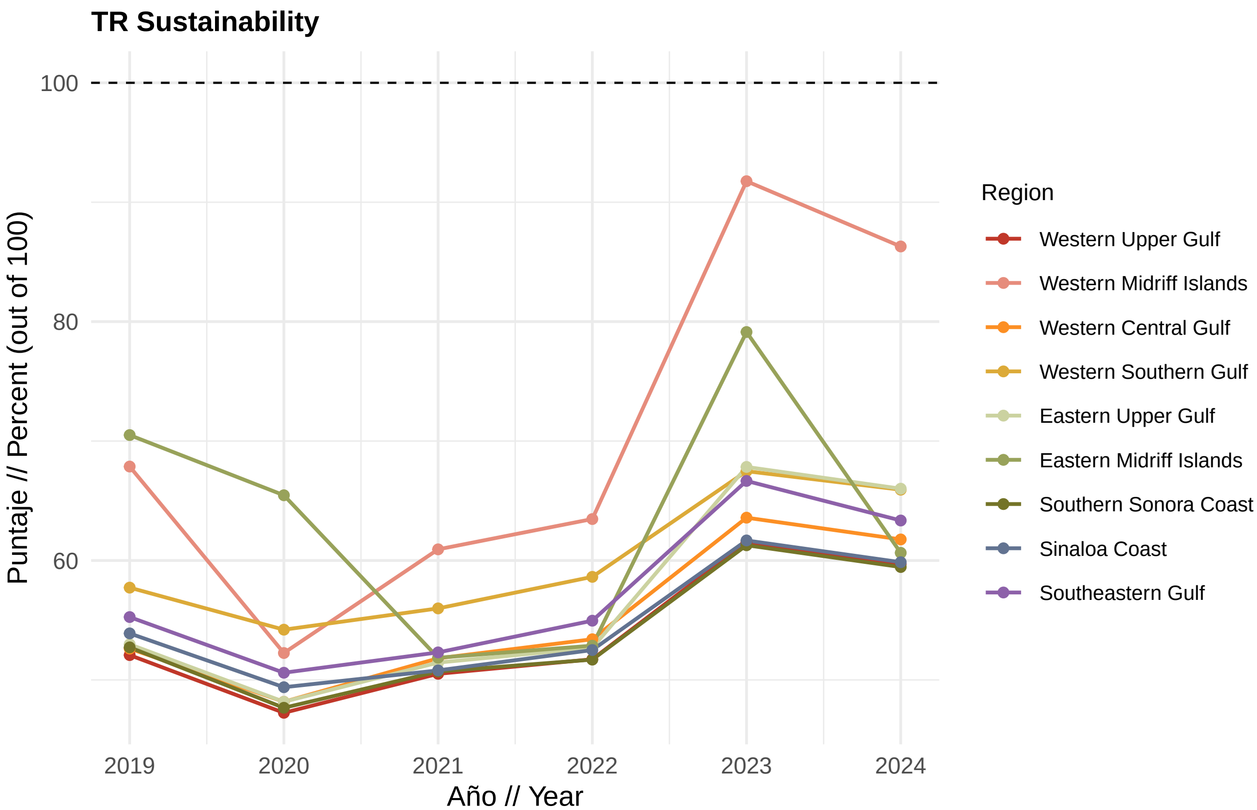
1.6.1 Sustainability component result

What it measures: Combined progress on sustainable certifications, tourism employment quality, and public investment in coastal management.

Data source: SECTUR, FEE, INEGI, and CONANP/SHCP as described above.

Method: Average of the certifications (hotel + Blue Flag), job quality, and investment scores.

$$\text{Sustainability}_{r,t} = \text{mean}\left(\text{Certifications}_{r,t},\ \text{JobQual}_{r,t},\ \text{Invest}_{r,t}\right)$$



1.6.1.1 Sustainability subcomponents

1.6.1.1.1 Certifications (hotel and Blue Flag combined)

What it measures: Share of large coastal hotels with sustainability certifications (Distintivo S, Green Key, or Green Globe) and share of urban tourist beach length under Blue Flag management.

What scores mean: Higher scores reflect greater adoption of recognized sustainability standards. For hoteliers and beach managers, rising scores indicate progress toward industry best practices; flat lines may signal certification barriers or limited program uptake.

Data source: SECTUR certification databases, Green Key and Green Globe member lists, INEGI DENUE accommodation units; FEE Blue Flag Mexico and INEGI urban beach linework.

Data processing: Hotels matched to DENUE by name/location; Blue Flag sites intersected with urban tourist beach buffers around Gulf centers.

Method: Hotel and Blue Flag scores are averaged for each region and year.

$$\text{Certifications}_{r,t} = \frac{\text{Hotels}_{r,t} + \text{BlueFlag}_{r,t}}{2}$$

1.6.1.1.2 Job quality in tourism

What it measures: Quality of formal employment in marine tourism sectors across maintained compensation, job formalization, and gender pay equity in formalization rates.

What scores mean: Higher scores indicate better-paid, more formalized, and more gender-equitable tourism jobs. Workers and employers can interpret sub-metrics separately: salary tracks compensation trends, formalization tracks contract coverage, and gender equity tracks whether women receive equal formalization rates as men.

Data source: INEGI Censos Economicos (2003, 2008, 2013, 2018, 2023), limited to marine tourism economic units, adjusted to constant 2023 pesos and coastal population multipliers.

Data processing: Census-year values are linearly interpolated to produce annual scores for 2019 to 2024 plotting. Coastal adjustments apply to Regions 3 and 4 and tourism-related sectors in Regions 1 to 9.

Method: Average of three sub-metrics (maintained salary, formalization, gender equity). Census-year scores are interpolated to annual values for plotting.

$$\text{JobQual}_{r,t} = \frac{\text{Salary}_{r,t} + \text{Formal}_{r,t} + \text{Gender}_{r,t}}{3}$$

1.6.1.1.3 Tourism management investment

What it measures: Federal investment in CONANP protected area programs relative to the 2012 to 2018 historical average, combined with the percentage of collected protected area fees returned from SHCP.

What scores mean: Higher scores indicate federal budgets and fee returns are at or above historical levels, supporting protected-area management that underpins sustainable tourism. Because budget programs are national, these scores apply uniformly across regions; fee return reflects treasury reinvestment policy.

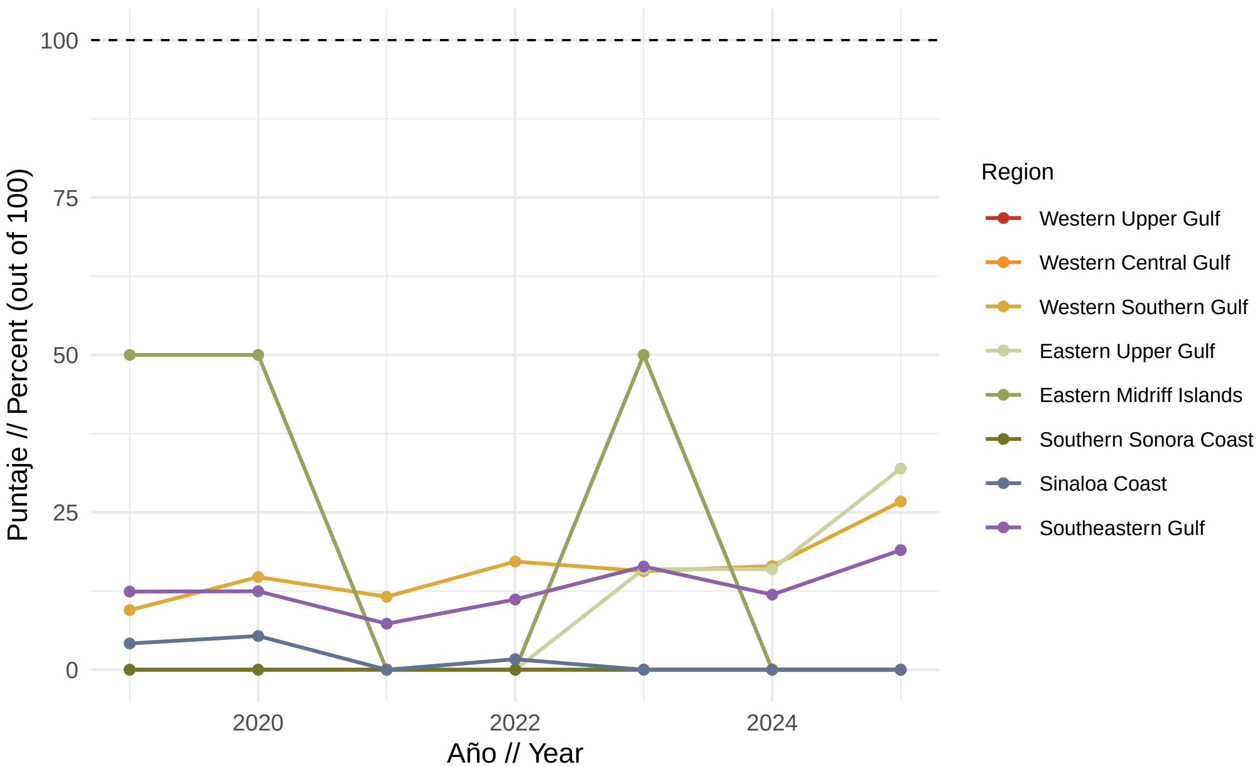
Data source: CONANP federal budget by program (v003 ANP management, s046 sustainable conservation, v005 protection and restoration); NOSSA/CONANP fee return data.

Data processing: Budget values converted to constant 2026 pesos; 2012 to 2018 mean used as reference for each program.

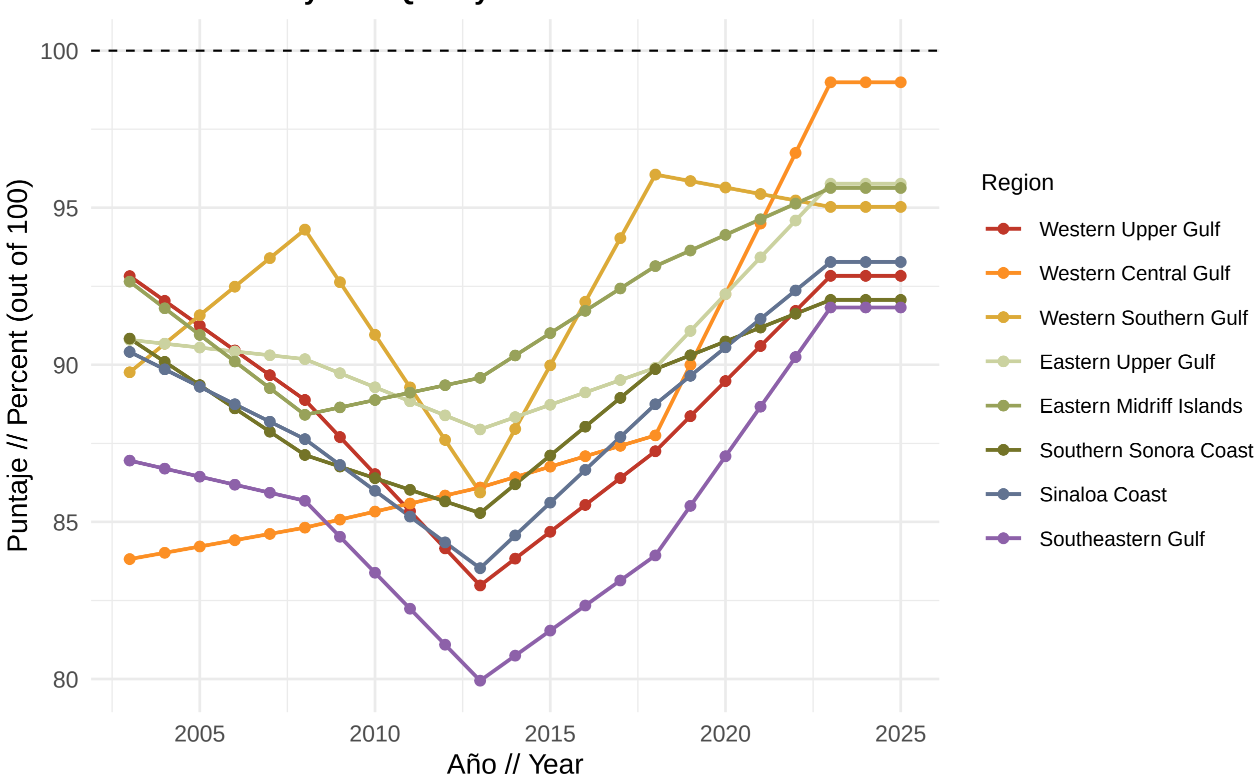
Method: Average of four program-level current status scores (three CONANP budget programs plus SHCP fee return). Budget programs use the 2012 to 2018 average as reference; fee return is expressed directly as a percentage.

$$\text{Invest}_{r,t} = \text{mean}\left(\text{ANP}_{r,t},\ \text{S046}_{r,t},\ \text{V005}_{r,t},\ \text{SHCP}_{r,t}\right)$$

TR Sustainability: Certifications (All)



TR Sustainability: Job Quality



1.6.1.1.4 Certification details

Hotel sustainability certifications

What it measures: Percentage of large hotels (100+ employees within 50 km of the Gulf coast) holding at least one sustainability certification (Distintivo S, Green Key, or Green Globe).

Data source: SECTUR Distintivo S registry, Green Key Mexico, Green Globe Americas member list, crosswalked to INEGI DENUE accommodation economic units.

Method: Certified large hotels divided by total large hotels, expressed as a percentage. Scores above 100 are capped at 100.

$$\text{Hotels}_{r,t} = \min\left(100,\ \frac{N_{r,t}^{\text{certified}}}{N_{r,t}^{\text{total large}}} \times 100\right)$$

Blue Flag beach certifications

What it measures: Share of urban/tourist beach length in each region that holds an active Blue Flag certification.

Data source: FEE Foundation for Environmental Education Blue Flag Mexico registry; INEGI coastline beach linework cropped to urban buffers around Gulf tourist centers.

Method: Length of Blue Flag certified tourist beaches divided by total tourist beach length, expressed as a percentage.

$$\text{BlueFlag}_{r,t} = \frac{L_{r,t}^{\text{blue flag}}}{L_{r,t}^{\text{tourist beaches}}} \times 100$$

1.6.1.1.5 Job quality details

Maintained salary

What it measures: Total compensation per paid worker (J000A / H010A) in marine tourism sectors relative to the 2003 to 2018 historical regional average.

Data source: INEGI Censos Economicos remuneration and employment variables for marine tourism units, in constant 2023 pesos.

Method: Current compensation per paid worker divided by the 2003 to 2018 regional average, capped at 100.

$$\text{Salary}_{r,t} = \min\left(100,\ \frac{(J000A_{r,t}/H010A_{r,t})}{\overline{J000A/H010A}_{r,\,\text{2003-2018}}} \times 100\right)$$

Formal/paid workers

What it measures: Share of tourism sector workers who are paid/formalized (H010A / H000A) relative to an ideal of 90% formalization.

Data source: INEGI Censos Economicos employment variables for marine tourism units.

Method: Current formalization rate divided by the 90% ideal, capped at 100.

$$\text{Formal}_{r,t} = \min\left(100,\ \frac{(H010A_{r,t}/H000A_{r,t})}{0.90} \times 100\right)$$

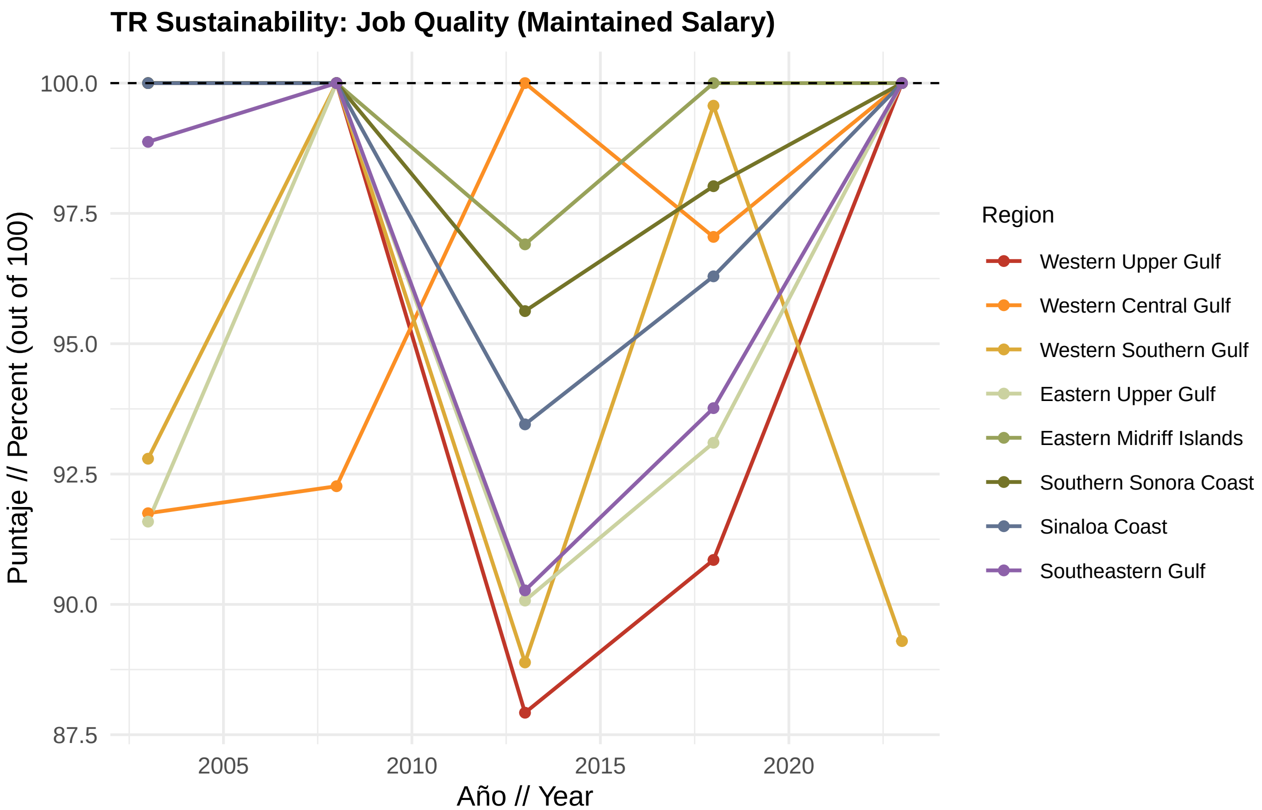
Gender pay equity (formalization equity)

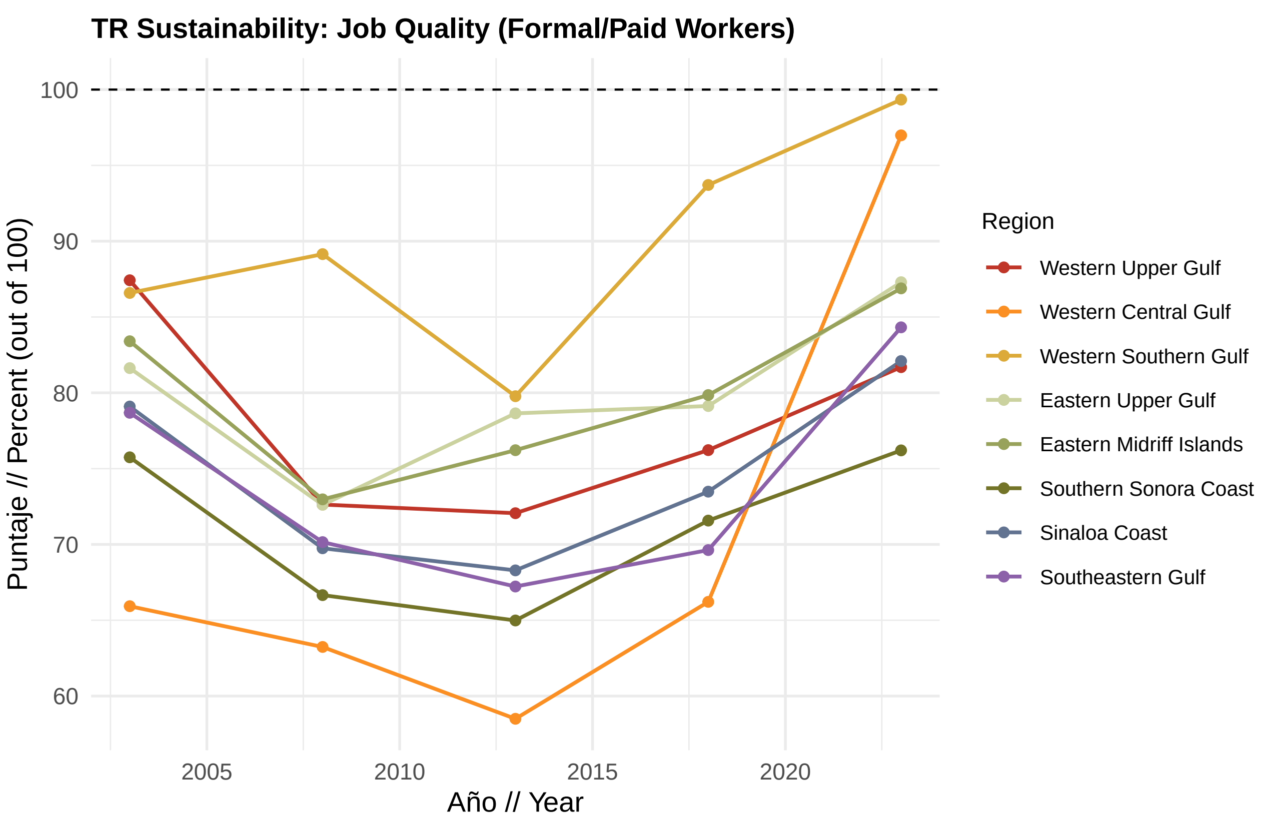
What it measures: Whether the share of formalized female workers equals or exceeds the share of formalized male workers in marine tourism.

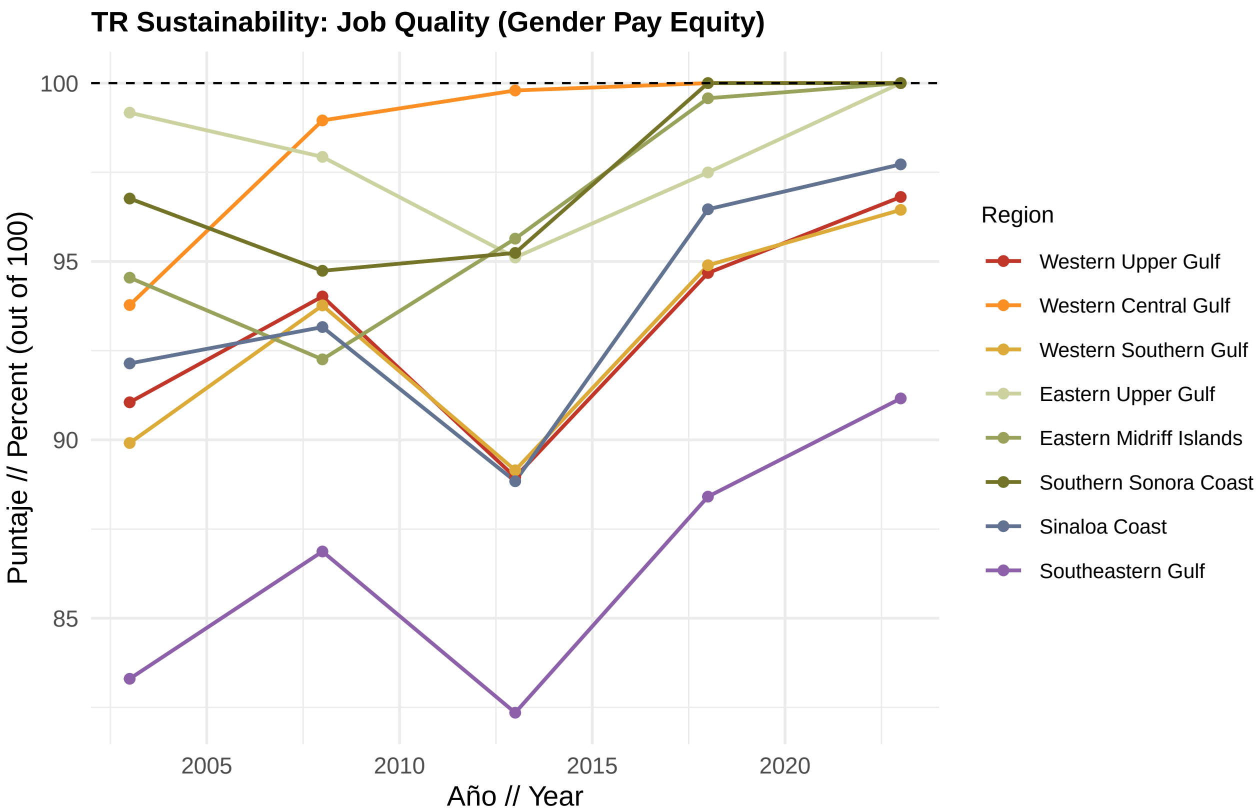
Data source: INEGI Censos Economicos gender-disaggregated employment (H010B/H000B for men, H010C/H000C for women).

Method: Ratio of female to male formalization rates. A ratio of 1 or greater scores 100; lower ratios are scaled proportionally.

$$\text{Gender}_{r,t} = \begin{cases} 100 & \text{if } \frac{(H010C/H000C)_{r,t}}{(H010B/H000B)_{r,t}} \geq 1 \\ \frac{(H010C/H000C)_{r,t}}{(H010B/H000B)_{r,t}} \times 100 & \text{otherwise} \end{cases}$$


TR Sustainability: Job Quality (Maintained Salary)
Puntaje // Percent (out of 100)
100.0
97.5
95.0
92.5
90.0
87.5
2005
2010
2015
2020
Año // Year
Region
Western Upper Gulf
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf


TR Sustainability: Job Quality (Formal/Paid Workers)
Puntaje // Percent (out of 100)
100
90
80
70
60
2005
2010
2015
2020
Año // Year
Region
Western Upper Gulf
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf



1.6.1.1.6 Investment program details

ANP management (v003)

What it measures: CONANP federal budget for Area Natural Protegida management (program v003) relative to the 2012 to 2018 historical average (constant 2026 pesos).

Data source: CONANP federal budget program data (`fed_budget_programas_importantes_2012_2026`).

Method: Current year budget divided by the 2012 to 2018 average, capped at 100. This score applies uniformly across all regions because federal budget is national.

$$\text{ANP}_t = \min\left(100,\ \frac{B_t^{\text{v003}}}{\overline{B^{\text{v003}}}_{\text{2012-2018}}} \times 100\right)$$

Sustainable conservation (s046)

What it measures: CONANP federal budget for sustainable conservation programs (s046) relative to the 2012 to 2018 historical average.

Data source: CONANP federal budget program data.

Method: Current year budget divided by the 2012 to 2018 average, capped at 100.

$$\text{S046}_t = \min\left(100,\ \frac{B_t^{\text{s046}}}{\overline{B^{\text{s046}}}_{\text{2012-2018}}} \times 100\right)$$

Protection and restoration (v005)

What it measures: CONANP federal budget for ecosystem protection and species-at-risk restoration (v005) relative to the 2012 to 2018 historical average.

Data source: CONANP federal budget program data.

Method: Current year budget divided by the 2012 to 2018 average, capped at 100.

$$\mathrm{V005}_t = \min\left(100,\ \frac{B_t^{\mathrm{v005}}}{\overline{B^{\mathrm{v005}}}_{\text{2012-2018}}} \times 100\right)$$

Returned fees from SHCP

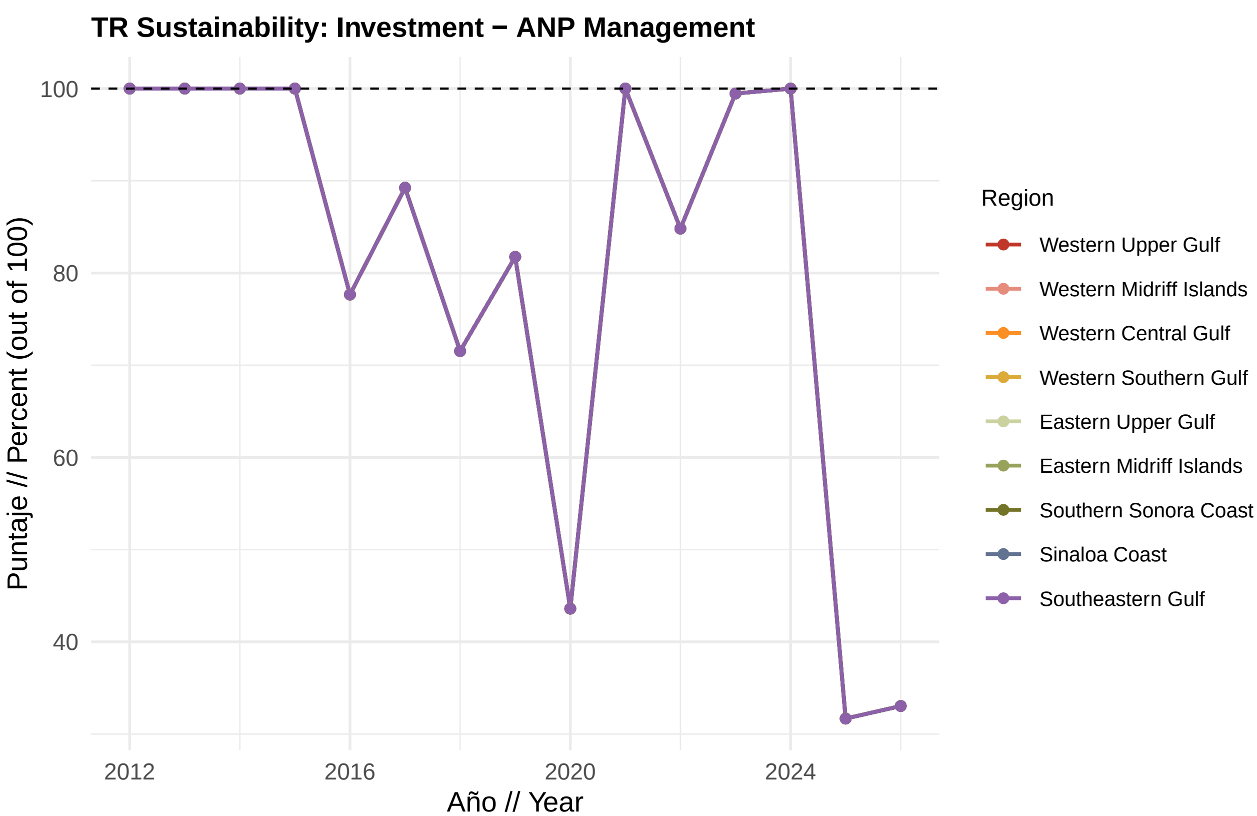
What it measures: Percentage of protected area entrance fees (cobro de derechos) collected by CONANP that are returned from the federal treasury (SHCP) for reinvestment in protected areas.

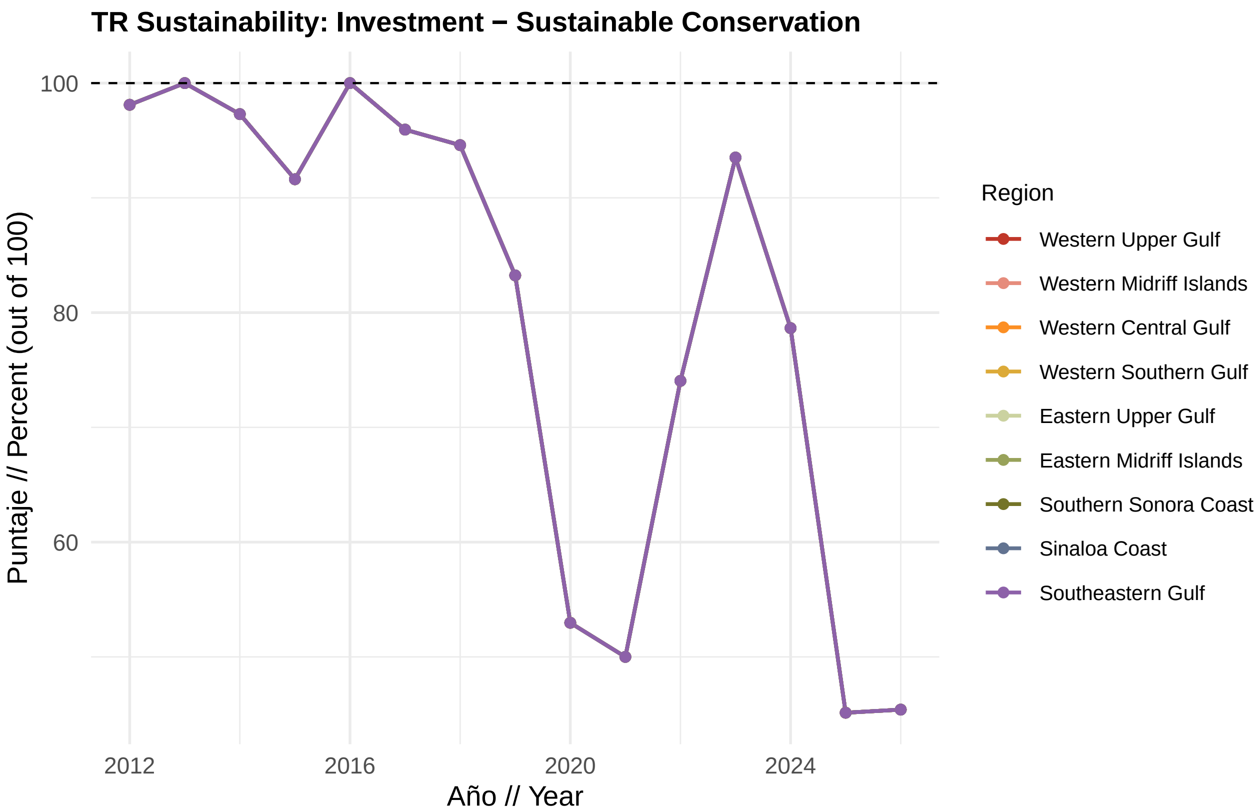
Data source: NOSSA/CONANP fee return records (`CONANP_fees_returned_2007_2026`).

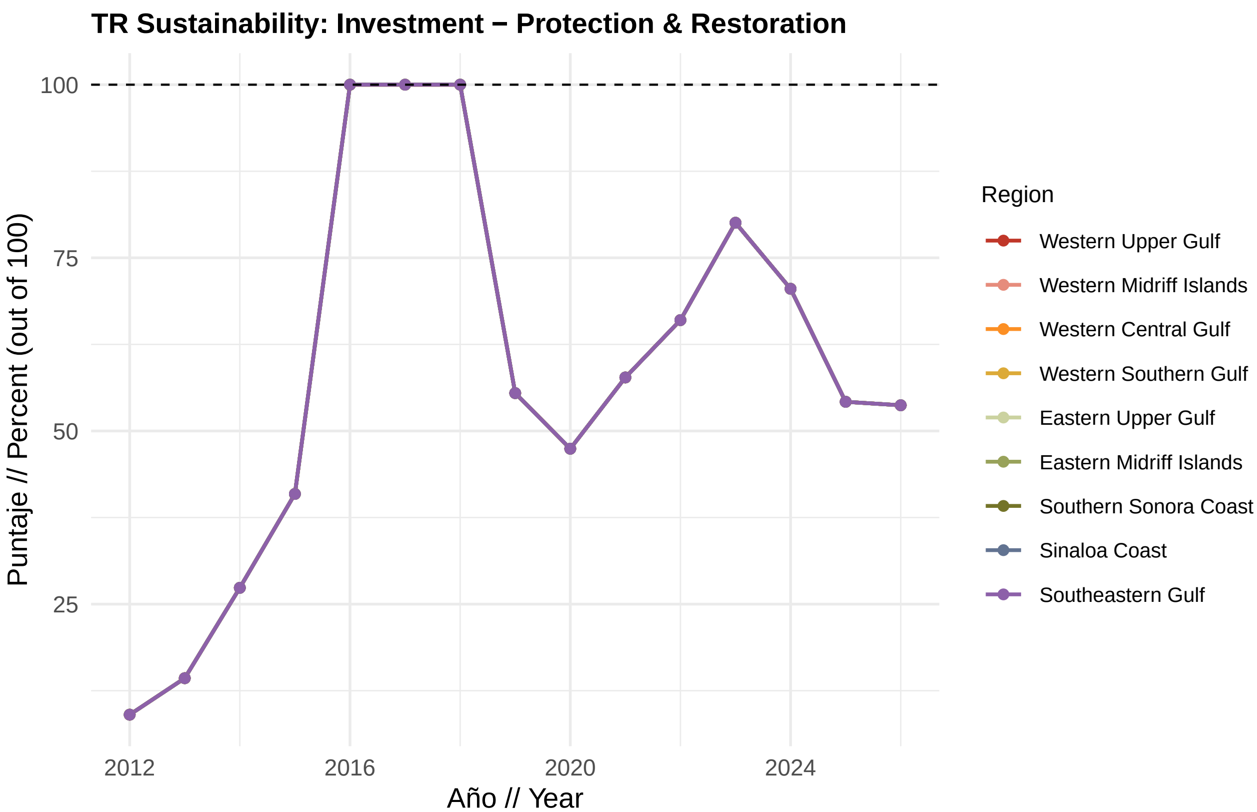
Method: The reported percentage returned is used directly as the current status score (100 = full return).

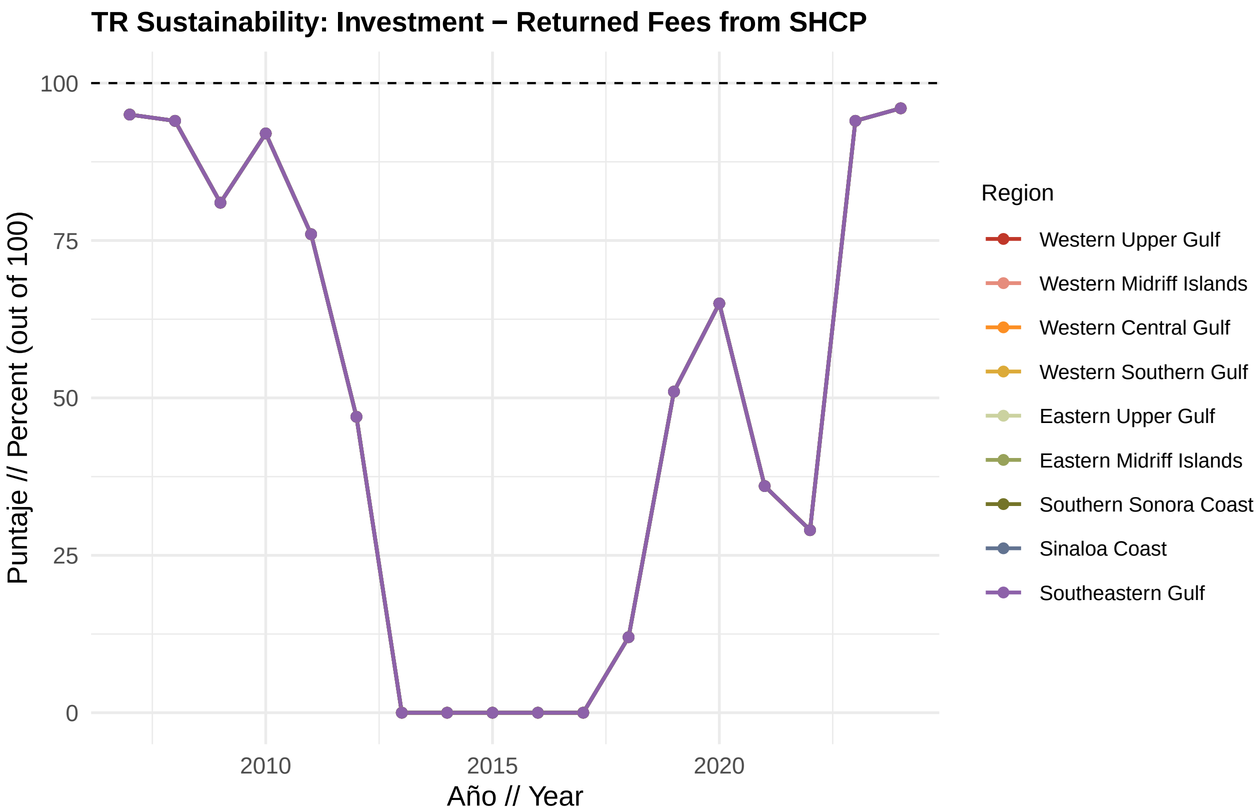
$$\mathrm{SHCP}_t = P_t^{\mathrm{return}} \times 100$$

where P_t^{return} is the fraction of collected fees returned in year t.




TR Sustainability: Investment − Sustainable Conservation
100
80
60
Puntaje // Percent (out of 100)
2012
2016
2020
2024
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf


TR Sustainability: Investment − Protection & Restoration
100
75
50
25
Puntaje // Percent (out of 100)
2012
2016
2020
2024
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf


TR Sustainability: Investment − Returned Fees from SHCP
Puntaje // Percent (out of 100)
100
75
50
25
0
2010
2015
2020
Año // Year
Region
Western Upper Gulf
Western Midriff Islands
Western Central Gulf
Western Southern Gulf
Eastern Upper Gulf
Eastern Midriff Islands
Southern Sonora Coast
Sinaloa Coast
Southeastern Gulf